

BIOS 794: Dimensions Reduction I

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Outline

Introduction

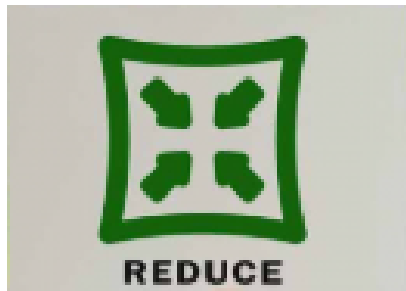
Local vs Global structure

- Local Structure

- Global Structure

- Methods for Local/Global

Local vs. Global Dimension Reduction



Why Dimension reduction?

- ▶ Dimension reduction refers to the process of reducing the number of random variables under consideration by obtaining a set of *principal variables*.
- ▶ It's a common step in data pre-processing to remove redundancy and irrelevant information, thereby helping in visualization, improving performance, and reducing computational costs.
 - ▶ **Feature Elimination** → simply eliminating irrelevant features.
 - ▶ **Feature Extraction** → transforming the data from a high-dimensional space into a space of fewer dimensions.

Why Dimension reduction?

- ▶ For our discussion here, we'll focus on feature extraction. Feature elimination, aka feature selection, has been and will continue to be discussed in other topics (LASSO, post-selection).
- ▶ Also, we won't discuss the following supervised methods in this section (all of which have feature extraction):
 - ▶ Linear Discriminant Analysis
 - ▶ Fisher's Linear Discriminant
 - ▶ Canonical Correlation Analysis

What Should a 2D Representation Preserve?

Suppose each face is described by hundreds of measurements (skin tone, nose size, eye spacing, lip shape, jaw width, ...). Some of these matter far more than others for telling faces apart.

We could have two goals when summarizing this data to 2D:

1. Faces that are close in 2D should have similar values on the features that differentiate faces the most — skin tone, nose size, male/female, ...
 - ▶ Close in this sense doesn't mean look-alike. Many small differences (eye spacing, lip shape, brow) add up, so two faces that match on the big features can still look quite different to a person.
2. Faces that look alike to a person should be close in 2D.
 - ▶ Example: if Face A's five most similar-looking faces are B–F, we want B–F to be A's neighbors in the 2D plot.

Local vs Global structure

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Local vs Global structure

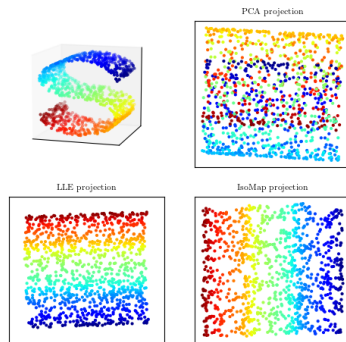
- ▶ Dimension reduction techniques can be categorized based on how they preserve the *local* and *global* structures of the data.
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- ▶ Techniques that emphasize local structure aim to ensure that nearby points in *the original space remain close to each other in the reduced-dimension space*.
- ▶ **Global structure** considers the broader relationships between data points that might be spread across the entire dataset.
- ▶ It aims to capture the data's overall trends and large-scale patterns.

Local structure

- ▶ Local structure focuses on capturing the fine-grained details and intricate patterns within clusters or groups of similar data points,
- ▶ preserving local structure is essential for retaining the inherent relationships between data points that are close to each other,
- ▶ techniques that emphasize local structure aim to ensure that nearby points in the original space remain close to each other in the reduced-dimensional space



Local structure example: Single-cell transcriptomic data

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- ▶ **Problem with global structure:**
 - ▶ Two subtypes of T-cells (say, activated vs. naïve T-cells) might differ by subtle expression differences in a handful of genes.
 - ▶ Global methods can miss these differences because they are focused on large-scale differences, and these genes might only explain a small fraction of total global variance.

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- ▶ **Problem with global structure:**
 - ▶ Global preservation could overemphasize large-scale variance driven by a few dominant features (like age or BMI).
 - ▶ The exact global distance between someone with diabetes and someone with asthma is less meaningful.

Local structure examples

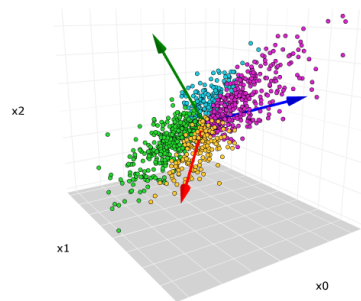
In public health genomics or precision medicine, we often care about:

- ▶ Identifying rare subgroups of cells or patients,
- ▶ Detecting early transitions in disease processes, or
- ▶ Understanding heterogeneity within a population.

All of these depend on preserving local relationships rather than the global geometry of the entire dataset.

Global structure

- ▶ Global structure considers the broader relationships between data points that might be spread across the entire dataset.
- ▶ It aims to capture the data's overall trends and large-scale patterns.
- ▶ Preserving global structure is important to understand the overall distribution and behavior of the data points, even when they are distant from each other in the high-dimensional space.



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 - ▶ A dimensionality reduction technique that emphasizes global structure could capture the major global trends, like countries with high/low GDP being located close to each other on one **DR variable** and/or countries with high/low violence close to each other on another **DR variable**.

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- ▶ **Problem with local space:**
 - ▶ In the original variable space, countries with similar socio-economic profiles might be scattered due to vast differences in isolated variables (e.g., Military expenditure, high for Israel, Saudi Arabia, and the US).

Global structure examples (cont)

- ▶ Other examples where the global structure is desired
 - ▶ **Financial Data Analysis:** Understanding the overall correlation structure between different stocks or assets is crucial in financial markets.
 - ▶ **Climate Studies:** Analyzing global weather patterns requires understanding large-scale relationships between different climate variables across regions.
 - ▶ **Social Network Analysis:** In studying social networks, understanding the overall community structure and how different groups are connected may be more important than focusing on individual connections.
 - ▶ **Medical Imaging:** In some medical imaging applications like MRI or CT scans, capturing the overall structure of the organ or tissue may be more relevant than focusing on local details.

Global and local structure

- ▶ One example where both global and local structure are desired is with single-cell RNA sequencing (scRNA-seq) data.
- ▶ **Local structure**
 - ▶ When analyzing scRNA-seq data, researchers are often interested in identifying subtle relationships between individual cells, such as small clusters representing specific cell subtypes in a developmental pathway.
 - ▶ In scRNA-seq, the local structure often captures biologically meaningful variations related to cell-type-specific gene expression patterns.
- ▶ **Global structure**
 - ▶ On the other hand, the global structure in scRNA-seq data might represent broader relationships between different cell types, tissues, or conditions.
 - ▶ Preserving global structure can be important when the focus is on understanding the broad developmental trajectories — e.g., stem cells → progenitors → differentiated cells.

Global and local structure

- ▶ The choice between local and global structure preservation depends on the characteristics of your data and the goals of your analysis.
- ▶ In some cases, preserving local structure might be more important, especially when dealing with clusters or non-linear relationships.
- ▶ In other cases, capturing global trends and distributions might take precedence, such as when visualizing the overall data distribution.
- ▶ A balanced approach might involve using a combination of techniques that capture both local and global structure.

Local structure methods

Examples of techniques that emphasize local structure include:

- ▶ **t-Distributed Stochastic Neighbor Embedding (t-SNE)**: t-SNE focuses on preserving pairwise similarities between data points, particularly emphasizing the local similarities in high-dimensional space. It is commonly used for data visualization and cluster analysis.
- ▶ **Locally Linear Embedding (LLE)**: LLE reconstructs each data point as a linear combination of its neighbors, emphasizing the local relationships. It aims to capture the underlying manifold structure.
- ▶ **Local PCA**: In cases where the data distribution is not globally linear, local PCA calculates principal components within small neighborhoods, capturing local linear relationships.

Global structure methods

Examples of techniques that emphasize global structure include:

- ▶ **Principal Component Analysis (PCA)**: PCA focuses on capturing the directions of maximum variance in the data. It emphasizes the global distribution of data points and projects them onto orthogonal axes that capture the largest variability.
- ▶ **Linear Discriminant Analysis (LDA)**: LDA aims to separate classes to maximize class separation. While it considers class relationships, it also captures global distribution differences.
- ▶ **Factor Analysis**: identify underlying latent variables (called factors) that explain the patterns of correlation among observed variables.

Methods good for both

Examples of techniques that emphasize both local and global structures in the data (all since 2016):

- ▶ **Uniform Manifold Approximation and Projection (UMAP)**: is a nonlinear dimension reduction technique that is especially useful for visualizing high-dimensional data.
- ▶ **LargeVis**: provides a balance between preserving distances between neighboring points and maintaining the overall relationships between clusters or groups, making it suitable for tasks such as visualizing complex datasets like large-scale single-cell RNA-sequencing data.
- ▶ **PaCMAP (Parameter-free Conformal MAPping)**: is a dimensionality reduction technique designed to preserve both local and global structures. It was introduced as an alternative to existing methods that require careful parameter tuning.

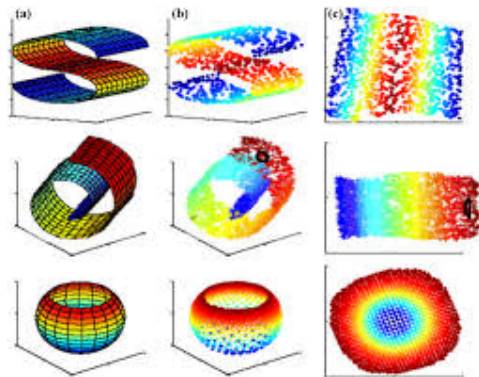
Global vs. Local Summary

Global methods (e.g., PCA, classical MDS)

- ▶ Preserve overall variance or large-scale distances.
- ▶ Capture broad trends or dominant directions in data.
- ▶ Often linear and reproducible.

Local methods (e.g., t-SNE, UMAP, Isomap)

- ▶ Preserve neighborhood relationships.
- ▶ Capture fine-grained structure (clusters, manifolds).
- ▶ Usually nonlinear and **non-invertible**.



When to Use Global vs. Local Methods

Global methods

- ▶ Used when you need transformations for **regression, prediction, or inference**.
- ▶ Transformations are linear and can be applied to new data.
- ▶ Examples: PCA, Factor Analysis, classical MDS.

Local methods

- ▶ Used for **visualization, clustering, or exploratory analysis**.
- ▶ Focus on local neighborhoods rather than overall distances.
- ▶ Examples: t-SNE, UMAP, Isomap.

Relation to Regression

- ▶ In regression or prediction, we prefer **global methods**:
 - ▶ Regression and causal modeling require global, interpretable, reproducible transformations.
 - ▶ Components are deterministic and can be reused for new observations.
 - ▶ Regression coefficients retain meaning on transformed scales.
 - ▶ Example: Principal Component Regression (PCR).
- ▶ Local methods (t-SNE, UMAP) are mostly for **exploratory visualization**:
 - ▶ Can show subgroups or nonlinear patterns.
 - ▶ But the axes have no fixed meaning—can't easily project new data.
 - ▶ Powerful for discovery, not for inference.

Examples in Public Health

- ▶ **Global method:** Summarizing dietary intake variables into a few “diet pattern” components via PCA, then using those in regression to predict BMI.
- ▶ **Local method:** Using t-SNE or UMAP to visualize clustering of individuals based on hundreds of biomarker measurements, to explore potential phenotypes.
- ▶ **Summary:**

Global methods for modeling and prediction.
Local methods for discovery and visualization.

Summary: Local vs. Global Methods

Goal	Preferred Type	Example Methods
Regression, prediction, inference	Global	PCA, MDS, Factor Analysis
Exploratory visualization	Local	t-SNE, UMAP, Isomap
Feature extraction (linear ML)	Global	PCA, ICA
Cluster or manifold discovery	Local	UMAP, PaCMAP